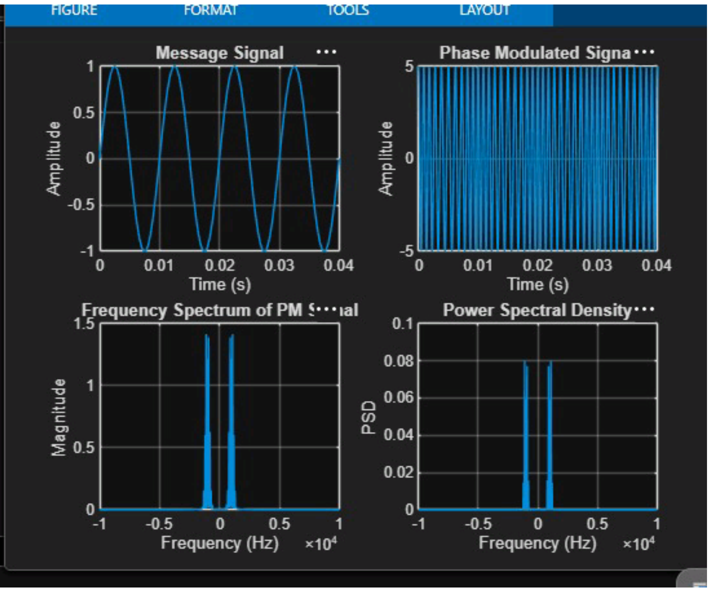
MATLAB Drive

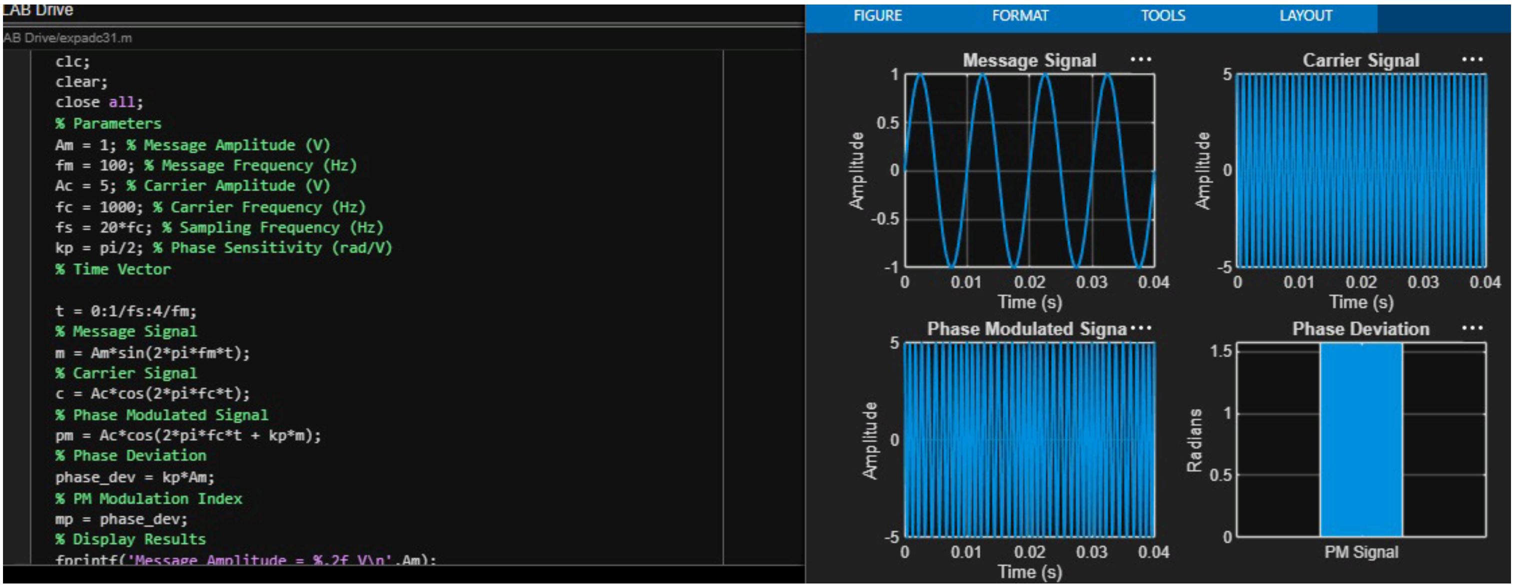
/MATLAB Drive/expadc32.m

```matlab
% Plotting
figure('Name','PSD Analysis of PM Signal','NumberTitle','off');
subplot(2,2,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,2);
plot(t,pm);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(2,2,3);
plot(f,PM_FFT);
title('Frequency Spectrum of PM Signal');
xlabel('Frequency (Hz)');
ylabel('Magnitude');
grid on;
subplot(2,2,4);
plot(f,PSD);
title('Power Spectral Density');
xlabel('Frequency (Hz)');
ylabel('PSD');
```

57 radians



```
Message Amplitude = 1.00 V
Message Frequency = 100 Hz
Carrier Amplitude = 5.00 V
Carrier Frequency = 1000 Hz
Phase Sensitivity = 1.57 rad/V
Phase Deviation = 1.57 radians
PM Modulation Index = 1.57
```


LAB Drive
AB Drive/expadc31.m
clc;
clear;
close all;
% Parameters
Am = 1; % Message Amplitude (V)
fm = 100; % Message Frequency (Hz)
Ac = 5; % Carrier Amplitude (V)
fc = 1000; % Carrier Frequency (Hz)
fs = 20*fc; % Sampling Frequency (Hz)
kp = pi/2; % Phase Sensitivity (rad/V)
% Time Vector
t = 0:1/fs:4/fm;
% Message Signal
m = Am*sin(2*pi*fm*t);
% Carrier Signal
c = Ac*cos(2*pi*fc*t);
% Phase Modulated Signal
pm = Ac*cos(2*pi*fc*t + kp*m);
% Phase Deviation
phase_dev = kp*Am;
% PM Modulation Index
mp = phase_dev;
% Display Results
fprintf('Message Amplitude = %.2f V\n',Am);
FIGURE
FORMAT
TOOLS
LAYOUT
Message Signal
Amplitude
Time (s)
Carrier Signal
Amplitude
Time (s)
Phase Modulated Signa
Amplitude
Time (s)
Phase Deviation
Radians
PM Signal